

**Exp No:1.c** Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

**Description:** Create small datasets for each type and explain their characteristics.

```
# Structured data example

structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})

print("Structured Data:\n", structured_data)


# Unstructured data example

unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,
or a video file."

print("\nUnstructured Data:\n", unstructured_data)


# Semi-structured data example (JSON)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

print("\nSemi-structured Data:\n", semi_structured_data)
```

**OUTPUT:**

Structured Data:

|   | ID | Name    | Age |
|---|----|---------|-----|
| 0 | 1  | Alice   | 25  |
| 1 | 2  | Bob     | 30  |
| 2 | 3  | Charlie | 35  |

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
}
```